

IT6 Final Drill

Enhancing a Flask Web Application with REST API Functionality

Due Date: May 31, 2025, 11:59PM

Total Points: 20

Instructions:

Students will enhance an existing Flask web application by adding a meaningful REST API feature that implements full CRUD (Create, Read, Update, Delete) functionality. The feature should align with and improve the original application's purpose.

Grading rubric (and requirements):

1. Project Selection (2 points)

- Choose a Flask Web application project with (ideally) clear documentation and modular structure, make sure you understand what it does and what feature you want to add into it, Fork the project, and proceed to the next step(s)
- Before proceeding, verify that your selected project isn't being used by anyone from your classmates. When certain it's unique, clone this repository, follow the instruction from there, apply your changes, and submit a pull request to the source project.

2. Feature Implementation (8 points)

- Identify a useful feature that would benefit from REST API implementation
- Implement complete CRUD operations for your chosen feature
- Ensure proper error handling and status codes (200, 201, 400, 404, etc.)
- Maintain clean code standards and proper documentation
- I highly suggest to set all your changes inside a branch so it's easy to see the changes.

3. Testing (5 points)

- Write unit tests with 100% coverage for your new feature
- Include both positive and negative test cases
- Use pytest or unittest framework
- Tests should verify all CRUD operations

4. Video Presentation (5 points)

- 5-7 minute technical presentation explaining:
 - The original application's purpose
 - Your added feature and its value

- Implementation details (routes, models, etc.)
- Testing approach
- Must show your face while presenting

Technical Requirements:

1. API Design:

- Use RESTful principles (proper resource naming, HTTP methods)
- Implement JSON request/response format
- Include proper status codes
- Add API documentation (Swagger/OpenAPI preferred)

2. Submission:

- Provide the original Repository
- Provide the URL of the cloned Github Repository with your changes (I suggest to make your new feature as new branch, and so that I can easily see your changes from the code of the original author)
- README.md explaining your changes
- YouTube/Google-video direct link to your presentation (MP4 format only)

Note: Plagiarism will result in zero points. All work must be original except for the base application you're enhancing. Video presentation is required in this take home drill (no video, no grade).